

"I feel recognized and developed a sense of belonging": Sustaining Volunteer Participation in Ancient Text Collation

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Abstract

This research explores factors that influence the sustainable volunteer participation in ancient text collation. We conducted in-depth interviews with 21 volunteers engaged in crowdsourced ancient text collation on an AI-facilitated platform. Through thematic analysis, we identified three types of individual motivation and three platform mechanisms that influenced their sustained participation. Our findings also highlighted the crucial role of affective state in maintaining long-term engagement. We further suggest mechanism design to enhance positive affective experiences through consistent communication, recognition, and autonomy, while minimizing negative affect by providing standardized feedback, preventing overload, and fostering belonging.

CCS Concepts

• **Human-centered computing** → **Empirical studies in collaborative and social computing**.

Keywords

Social participation; crowdsourcing; social and crowd computing; cultural heritage; ancient text digitization; sustainable volunteer participation; citizen science

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1 Introduction and Related Work

Crowdsourcing and citizen science have emerged as powerful approaches to accomplish complex tasks by harnessing the collective efforts of volunteers. Particularly, the significance of social participation has been widely acknowledged in preserving cultural heritage[5]. While crowdsourcing platforms have been successful in attracting initial volunteer participation, sustaining long-term engagement remains a challenge in knowledge-intensive domains like ancient text collation. Originally preserved on paper, ancient Chinese texts are difficult to access and maintain, which limits their availability to the broader public[18]. Ancient text collation, involving transcription, structural and punctuation tagging, and traditional-to-simplified character conversion, transforms physical manuscripts into accessible and readable digital formats[3]. These tasks are usually monotonous, repetitive, and relies on specialized expertise. The automation of traditional collation processes through AI has reduced barriers to entry, enabling broader participation from volunteers with diverse expertise levels. However, this technological advancement makes sustainable engagement even more critical. Volunteers need time to develop the specialized skills and cultural knowledge necessary for accurate collation work, making short-term participation insufficient for maintaining quality standards. Successful large-scale digitization projects depend on sustainable participation to build and retain a skilled volunteer community that can ensure consistency, mentor newcomers, and maintain the necessary cultural sensitivity.

To effectively harness volunteer participation, it is essential to understand what motivate them to contribute. Motivations for participation vary among different types of volunteers[2]. Previous research has extensively examined volunteers' motivations for participating in social-purpose work across various contexts. Volunteers may participate in crowdsourcing more out of the well-being of others than for personal benefit[20]. Kobayashi et al. categorized volunteers' motivation into personal and social motivations[16]. Personal motivations include skill fitness, task autonomy, task identity, fun, and interest, while social motivations encompass community identification, social contribution, and more. Research also indicates that these motivations are temporal and dynamic [27], with different

value orientations impacting participation behaviors [15]. Volunteers' motivations are driven by external factors, cognitive value, emotional perception, and task factors [31]. Crowdsourcing platforms should design to motivate volunteers' contributions, across the themes of task visibility, goals, feedback, and rewards [26]. Parallel to motivation studies, considerable research has focused on platform and mechanism design to promote volunteer participation. Some research has explored mechanisms to enhance volunteers' willingness to participate by designing simple tasks and providing platform support [6, 8, 9, 24], while others have investigated strategies to improve participation quality through capability-based task allocation [13] and positive peer feedback [7, 14, 22]. Additional studies have examined platform optimization for volunteer management and coordination [4, 19] and effective communication between project administrators and volunteers [25, 28, 30]. Research on sustained participation has highlighted the importance of collaborative culture [23, 29], platform support for volunteer interaction [12], and fostering a sense of belonging [11].

Despite these valuable contributions, there remains a gap in understanding factors influencing sustainable participation in specialized knowledge domains like ancient text collation. Most existing research has either focused on general crowdsourcing contexts or examined motivational factors and platform mechanisms in isolation, without sufficient consideration of their combined impact on long-term participation. Additionally, while research acknowledges that motivations evolve over time, limited attention has been paid to identifying the specific combination of factors that sustain volunteer engagement beyond initial participation in knowledge-intensive cultural heritage work.

Therefore, this study explores the research question: *What factors influence sustainable volunteer participation in ancient text collation and how?* By examining both individual factors and platform mechanisms in this specialized context, our research aims to provide insights into the complex dynamics that support long-term volunteer engagement in cultural heritage preservation efforts. To address the research question, we conducted in-depth interviews with 21 volunteers engaged in an ancient text collation crowdsourcing program. Through thematic analysis, we identified three types of individual motivations and three platform mechanisms that influence volunteers' sustained participation. The findings highlight the significance of affective states in sustained participation—highly engaged volunteers were primarily driven by affective motivations and gained positive affect during continuous participation. We provide suggestions on platform mechanism design to enhance positive affective experiences while minimizing negative ones.

2 Methodology

2.1 Research Setting

To support the crowdsourced collation of ancient texts, the Research Center for Digital Humanities at Peking University (PKUDH) has designed the “Shidian Guji” platform (“Shidian” hereafter) in collaboration with ByteDance CSR¹. The platform uses AI to facilitate the collation of ancient texts by automating tasks like transcription

from scanned images, structural and punctuation tagging, named entity tagging, etc. While the platform enabled public to contribute to the intelligent collation of ancient texts through crowdsourcing, the vast majority of the collated, reader-accessible texts on the platform have been prepared by volunteers organized by PKUDH. PKUDH employs a project-team approach to organize volunteers, assigning each project to a compact team composed of several volunteers and one or two on-site deeply involved volunteers acting as project leaders. Since the platform was developed in 2022, PKUDH has mobilized over 600 volunteers to collate over 19,000 ancient books on Shidianguji for public access. It offers a great case to investigate volunteer engagement and sustainability.

2.2 Participants and Data Collection

Between December 2024 and March 2025, we conducted interviews with volunteers engaged in ancient text collation on Shidian. For participant selection, we adopted a combination of purposeful sampling and snowball sampling. We first distributed a screening survey in the volunteers' social-media group chat to identify participants with varying lengths of participation and roles. We also asked the initial participants of the interview to recommend other eligible volunteers, particularly those with shorter engagement. In total, we recruited 21 volunteers. The sample covers all types of roles and responsibilities and represents a balanced distribution in duration of engagement, with 10 participants engaged less than 6 months, and 11 participants engaged 6 months or longer. Table 1 summarizes the detailed information of the interview participants:

We employed semi-structured, one-on-one interviews to gain an in-depth understanding of the participants' perspectives. Each interview lasted between 30 to 60 minutes. The interview questions covered a range of topics, including participants' motivations for involvement, their responsibilities, experiences during participation, and the challenges faced during the process of ancient text collation. These volunteers varied in levels of engagement and undertook different types of roles and responsibilities, which allowed us to gain a more comprehensive understanding of participation patterns and the factors that influence sustained involvement.

2.3 Data Analysis

We conducted thematic analysis to inductively code the interview transcripts. At the beginning, we individually open coded different sets of transcripts. We then merged and compared the codes, grouped similar ones, and assigned them identical labels. Throughout the coding process, we discussed on a weekly basis to establish consistent understanding of codes and resolve any uncertainty and disagreement. Following the generation of preliminary codes, we employed an iterative process of theme identification, review, and refinement to inductively derive distinct and non-overlapping themes. In terms of individual motivations, we identified three major themes: self-development motivation, prosocial motivation, and affective motivation. In terms of platform mechanisms, we identified the task allocation mechanism, review mechanism, and team collaboration mechanism. These two aspects together contribute to sustained participation.

¹ByteDance is a Chinese internet technology company. ByteDance's Corporate Social Responsibility (CSR) activities in China include initiatives focused on social good, environmental sustainability, and community engagement.

Table 1: Participants' Information

No.	Gender	Roles and responsibilities	Duration of engagement
P01	Female	Volunteer management, proofreading and annotation, academic research	6 months
P02	Male	Volunteer management, proofreading and annotation, data cleaning, academic research	24 months
P03	Male	Data cleaning	2 months
P04	Male	Proofreading and annotation, data cleaning, volunteer management, academic research	12 months
P05	Male	Proofreading and annotation	1 month
P06	Male	Proofreading and annotation	12 months
P07	Female	Proofreading and annotation, volunteer management	24 months
P08	Male	Proofreading and annotation	4 months
P09	Male	Proofreading and annotation	12 months
P10	Female	Proofreading and annotation	2 months
P11	Female	Proofreading and annotation	2 months
P12	Male	Proofreading and annotation	Less than 1 month
P13	Male	Proofreading and annotation, volunteer management	24 months
P14	Male	Data cleaning, academic research	6 months
P15	Male	Proofreading and annotation, volunteer management	18 months
P16	Male	Proofreading and annotation, volunteer management, data cleaning, academic research	24 months
P17	Male	Proofreading and annotation, volunteer management	24 months
P18	Female	Proofreading and annotation	Less than 1 month
P19	Female	Proofreading and annotation	Less than 1 month
P20	Female	Proofreading and annotation	Less than 1 month
P21	Male	Proofreading and annotation	Less than 1 month

Note. Proofreading and annotation includes proofreading the OCR results, annotation of structural element, punctuation, and named entity, etc. Data cleaning includes text segmentation, image cropping, text-image alignment, etc.

3 Findings

3.1 Individual Motivations Influencing Sustainable Participation

The analysis revealed three types of individual motivations influencing sustainable participation: self-development motivation, prosocial motivation, and affective motivation. Unlike one-time extrinsic motivations such as payment, these motivations represent deeper, more enduring drivers that foster long-term commitment and sustained engagement over time. Self-development motivation refers to the drive to participate in activities based on personal growth goals, interests, or skill improvement. Prosocial motivation refers to doing things in order to help others or the society, such as contributing to the cause of culture inheritance. Affective motivation means being driven mainly by emotional satisfaction, such as enjoyment, a sense of accomplishment or belongings.

In terms of self-development motivation, participants with sustained engagement viewed the experience as aligned with their long-term career development goals rather than merely short-term gains. The majority of participants were undergraduate or graduate students currently enrolled in universities, who are in the process of choosing their future career paths. Some voluntarily participated as a way to accumulate work experience, and thus enhancing their competitiveness in the job market. For example, participant P10 mentioned, “*I’d like to be an editor. As merely studying from books can’t get any practice experience, I chose to volunteer to improve my editorial skill.*” Unlike participants seeking one-time benefits, such

as internship certificate, those with strong self-development motivation saw it as an ongoing opportunity for professional growth, which contributed to their continued participation.

Regarding prosocial motivation, participants’ sustained engagement is primarily driven by their enduring identification with the broader social and cultural goals of the platform. Most participants spoke highly of the platform for leveraging public participation, making it more accessible than traditional collation methods, and playing a key role in promoting traditional culture. As participant P12 noted, “*This approach expands the scope of ancient text collation through digital approach which I think is aligned with broader trends.*” This alignment between personal values and project mission created a strong foundation for long-term commitment, as volunteers continued to see their work as meaningful contributions to cultural preservation over extended periods.

In terms of affective motivation, many participants with sustained engagement reported experiencing ongoing affective rewards from their participation. Many participants contributed to the platform due to their interest in ancient texts, perceiving collation not as a task, but as a form of leisure or personal interest. Further, to some participants, affective motivation is closely related to their prosocial motivation. Participants obtained a continuous sense of attainment believing that they are contributing to a greater cultural cause. This recurring emotional satisfaction motivated their continuous contribution over time. As participant P14 stated, “*I’m quite satisfied with my current experience and I’m also genuinely excited to see how the platform continues to grow.*” This forward-looking

enthusiasm indicates how affective motivation supports not only current participation but also anticipation of future engagement.

3.2 Platform Mechanisms Influencing Sustainable Participation

We identified three platform mechanisms that influence sustainable participation: the task allocation mechanism, review mechanism, and team collaboration mechanism. We found that incentive mechanism had minimal effect on sustained participation, whereas these three mechanisms influenced sustainable participation through affective impact on volunteers. They either created positive affective experiences like recognition and belonging, or generated negative affective state, such as frustration and stress, that led volunteers to consider terminating their participation.

In terms of task allocation mechanism, participants were allowed to self-select tasks, which enhanced autonomy and flexibility, enabling them to pause and resume work at their convenience, thereby encouraged more sustained participation. This flexibility proved particularly important for maintaining long-term engagement, as volunteers could adapt their participation to varying life circumstances without having to withdraw completely. However, this autonomy-based approach also introduces risks that can undermine sustainable participation. It can lead to unfulfilled commitments, as individuals may claim multiple assignments without completing them, creating bottlenecks that impede others and delay the entire project. For instance, Participant P05 observed that many volunteers claimed a large number of tasks they were interested in, but failed to follow through because of other things at hand.

The review mechanism encompasses both efficiency and the establishment of standards. While reviews by project leaders help ensure quality, limited efficiency has raised concerns about sustaining participation. For instance, Participant P06 remarked, *“Sometimes it takes a day or two to receive feedback. And by then, other tasks or distractions may have come up, making it difficult for volunteers to respond promptly to their mistakes.”* These delays create discontinuity in the volunteer experience, making it harder to maintain consistent engagement over time. As the volunteer and task scale expands, standardizing the review mechanism also becomes an important issue for sustained volunteer participation. Inconsistencies in review standards may lead to frustration that erodes long-term commitment. For example, Participant P10 stated, *“I think the review standards differ between reviewers. One reviewer once told me that a basic level of quality was sufficient for a certain task, while another reviewer seemed to hold much higher standards.”* This unclear standard dampened her enthusiasm because she felt uncertain about how to carry out future tasks.

The team collaboration mechanism includes communication among volunteers and with project leaders, as well as the overall team atmosphere. Most participants emphasized that a positive, supportive environment was key not only to initial satisfaction but also to sustained engagement. For example, participant P01 highlighted the communication style of her project leader as especially effective for fostering long-term engagement: *“When one’s work has issues, our leader would say, ‘Thanks for your effort. This part is great, and a small revision in this section would improve it.’ His way*

of communication gradually built trust within the team. I feel recognized and developed a sense of belonging.” This ongoing positive reinforcement created emotional bonds and a sense of belonging that motivated participants to continue contributing even when facing challenges or competing priorities.

4 Conclusion and Discussion

In this study, we revealed the key factors underlying sustained volunteer participation and elucidated the ways in which they influence volunteer engagement over time. The findings also revealed the significance of affective states in sustained participation—not only the affective motivations, but also the mechanisms that have an affective impact on them. Based on these findings, we derived implications for mechanism design.

4.1 The Importance of Affective States in Sustained Participation

Previous literature categorized volunteers’ motivation in social-purpose work into personal and social motivations [16]. We found that volunteers who have longer duration of engagement are mainly driven by affective motivations, including personal motivations such as sense of accomplishment (or task identity [16]), fun, interest [1], and social motivations like encouragement from the community, and community identification [17]. Most of the highly engaged volunteers are passionate history and culture enthusiasts. For them, the prosocial motivation, which is to contribute to the preservation and promotion of traditional culture, can also turn into a strong sense of accomplishment and community identification to motivate their continued participation.

Apart from the positive affective factors, volunteers may also face excessive pressure and feelings of frustration, which may cause them to consider terminating participation. For instance, balancing academic obligations with volunteer responsibilities often poses a significant challenge. Participant P13, a team leader, described experiencing moments when up to eight team members sought assistance simultaneously outside of working hours. The demands of staying up late and managing time commitments became a major source of stress, prompting serious thoughts about quitting the project. Therefore, the platform mechanisms should pay more attention to the affective states in order to sustain volunteer participation. It is important to design mechanisms towards positive affective factors, such as sense of accomplishment, belonging, and enjoyment, and to avoid negative affective factors, such as stress and frustration.

4.2 Mechanism Design for Sustainable Participation

Previous studies have shown that volunteer engagement is closely tied to the quality of their relationship with project leaders [21]. Establishing regular communication channels between volunteers and project coordinators plays a crucial role in strengthening emotional bonds and fostering team cohesion, thereby promoting sustained participation over time. To cultivate a sense of belonging and recognition, it is essential to identify volunteers’ motivations and expectations during the recruitment and onboarding phases—an approach that aligns with insights from existing CSCW research

[15]. In terms of task allocation, setting appropriate workload limits based on volunteers' past performance helps prevent burnout while also encouraging a sense of progress and accomplishment. For specialized tasks such as classical text collation, involving volunteers in internal discussions allows them to contribute to the refinement of task standards. Providing a degree of autonomy, rather than enforcing rigid procedural adherence, tends to enhance their sense of ownership and deepen their commitment. Although previous research emphasizes the effectiveness of incentive mechanisms like leader boards [10], our findings suggest that such strategies may be less useful in ancient text collation crowdsourcing, where social value is more prioritized.

To mitigate negative affect, the design of project mechanisms should aim to reduce stress, frustration, and perceptions of unfairness. One major source of volunteer dissatisfaction is inconsistent or delayed feedback. Implementing transparent and standardized evaluation criteria can reduce confusion and the perception of subjectivity in assessments. Additionally, optimizing the review process to ensure timely feedback can help volunteers feel that their efforts are acknowledged and valued. Limiting excessive workloads and clarifying expectations can also alleviate stress, especially for volunteers balancing participation with other responsibilities.

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